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- Cytokinin can break the dormancy of certain light-sensitive seeds like tobacco and induces seed germination.
- Cytokinin promotes the growth of lateral bud in the presence of apical bud.
- Application of cytokinin delays the process of aging by nutrient mobilization. It is known as **Richmond Lang effect**.
- Cytokinin (i) increases rate protein synthesis (ii) induces the formation of inter-fascicular cambium (iii) overcomes apical dominance (iv) induces formation of new leaves, chloroplast and lateral shoots.
- Plants accumulate solutes very actively with the help of cytokinins.

15.2.4 Ethylene (Gaseous Phytohormone)

Almost all plant tissues produce ethylene gas in minute quantities.

1. Discovery

In 1924, **Denny** found that ethylene stimulates the ripening of lemons. In 1934, **R. Gane** found that ripe bananas contain abundant ethylene. In 1935, **Cocken** *et al.,* identified ethylene as a natural plant hormone.

2. Occurrence

Maximum synthesis occurs during climacteric ripening of fruits (*see* Box info) and tissues undergoing senescence. It is formed in almost all plant parts like roots, leaves, flowers, fruits and seeds.

3. Transport in plants

Ethylene can easily diffuse inside the plant through intercellular spaces.

4. Precursor

It is a derivative of amino acid methionine, linolenic acid and fumaric acid.

5. Bioassay (Gas Chromatography)

Ethylene can be measured by gas chromatography. This technique helps in the detection of exact amount of ethylene from different plant tissues like lemon and orange.

6. Physiological Effects

- Ethylene stimulates respiration and ripening in fruits.
- It stimulates radial growth in stem and root and inhibits linear growth.
- It breaks the dormancy of buds, seeds and storage organs.
- It stimulates formation of abscission zone in leaves, flowers and fruits. This makes the leaves to shed prematurely.
- Inhibition of stem elongation (shortening the internode).
- In low concentration, ethylene helps in root initiation.
- Growth of lateral roots and root hairs. This increases the absorption surface of the plant roots.
- The growth of fruits is stimulated by ethylene in some plants. It is more marked in climacteric fruits.
- Ethylene causes epinasty.

Agricultural role

- Ethylene normally reduces flowering in plants except in Pine apple and Mango.
- It increases the number of female flowers and decreases the number of male flowers.
- Ethylene spray in cucumber crop produces female flowers and increases the yield.